Too Cool To Do Compilers





About Me













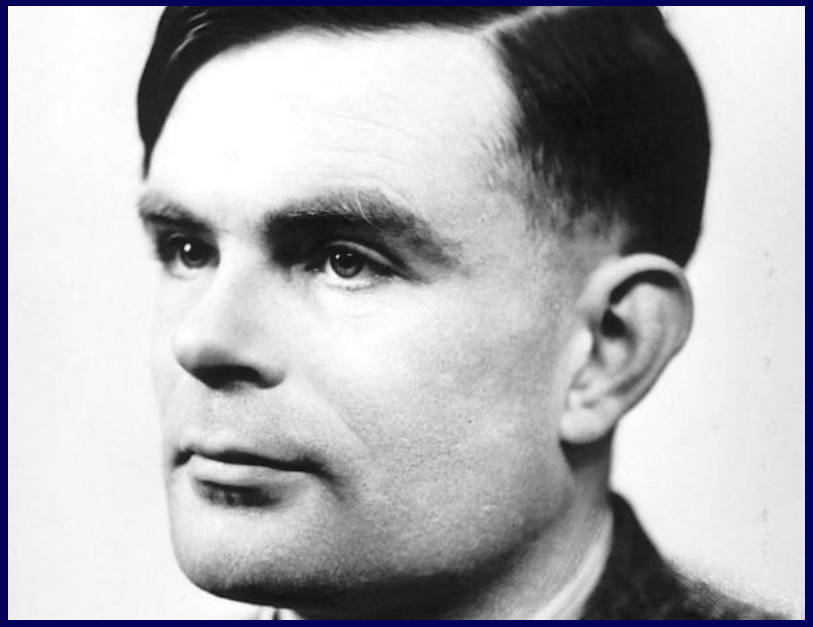
PL

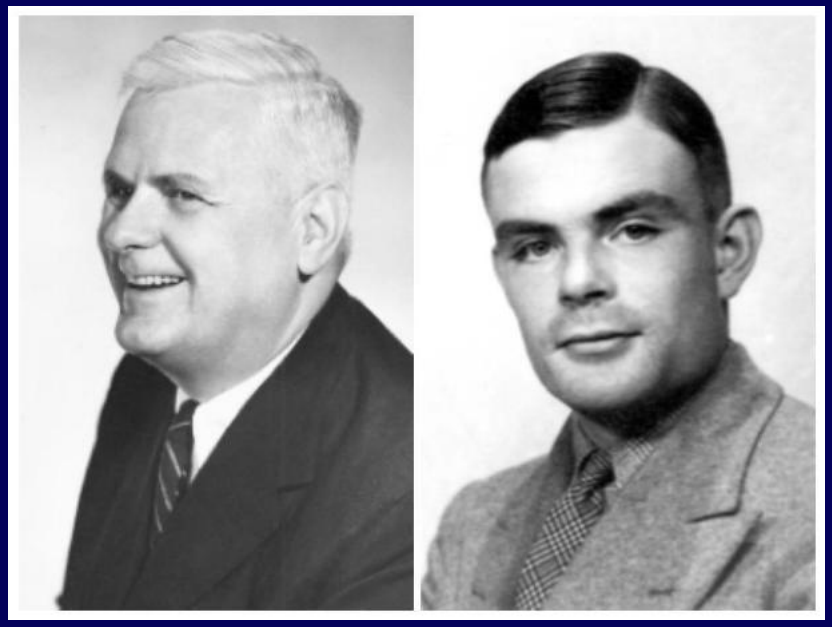


- IDE VIM, IDEA, clion, Pycharm, VSC, VS, DevCPP
- Compiler gcc/g++, clang, javac, rustc
- Linker ld, lld, ms link
- ToolChain mingw, rust toolchain, clang toolchain
- Build System cargo, maven, gradle, cmake, xmake, qmake, make



PL What is Computation?





什么是计算？

PL

Programming Paradigms

- Imperative

  OOP (Object-Oriented Programming)

```
void quicksort(vector<int>& a, int l, int r) {
    if (l >= r) return;
    int i = l, j = r, x = a[(l + r) / 2];
    while (i <= j) {
        while (a[i] < x) ++i;
        while (a[j] > x) --j;
        if (i <= j) swap(a[i++], a[j--]);
```

$$F(L) = \begin{cases} [], & L = [] \\ F([a \mid a \in L', a < x]) + [x] + F([a \mid a \in L', a \geq x]), & L = x : L' \end{cases}$$

  FP (Functional Programming)

```
quicksort :: Ord a => [a] -> [a]
quicksort [] = []
quicksort (x:xs) =
    quicksort [a | a <- xs, a < x]
    ++ [x] ++
    quicksort [a | a <- xs, a >= x]
```

  LP (Logic Programming)

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```
bool check(int x) {
    return x + 1 > x;
}
int main(int a, char **b) {
    return check(a);
}
```

```
"check(int)":
        mov     eax, 1
        ret
"main":
        mov     eax, 1
        ret
```

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```
// Common Subexpression Elimination
void foo2(int* a, int n, int x, int y) {
    for (int i = 0; i < n; ++i) {
        int t1 = x + y;
        int t2 = t1 * t1;
        a[i] = t2;
    }
}
```

```
// Original
void foo(int* a, int n, int x, int y) {
    for (int i = 0; i < n; ++i) {
        a[i] = (x + y) * (x + y);
    }
}
```

```
void foo3(int* a, int n, int x, int y) {
    int t1 = x + y;
    int t2 = t1 * t1;

    for (int i = 0; i < n; ++i) {
        a[i] = t2;
    }
}
```

Cool To Do Compilers

- 计算机本科课程必修
- ？？！ 正则表达式？？！
- 我的循环能不能被并行化?
- 嵌入式经常见到的 volatile 究竟是什么?
- 为什么函数式大量使用递归而不担心性能和溢栈?
- 为什么在 C++ 和 rust 语言中如此推崇不可变?
- 编译器能不能静态判定一个程序是否会死循环?
- 我们能期待编译器做什么，不能期待做什么?

> Some people, when confronted with a problem, think
> “I know, I'll use regular expressions.” Now they have two problems.
>
> — Jamie Zawinski

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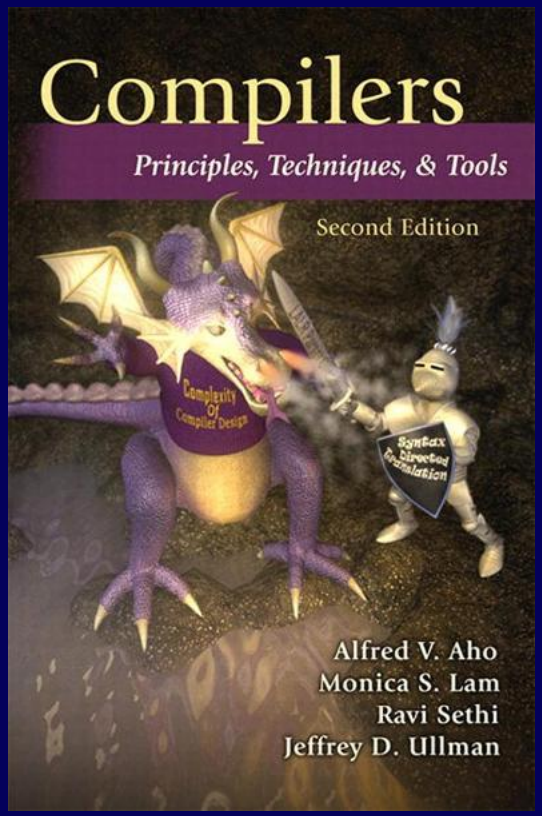
- Compiler Optimization
- Program Analysis
- Programming Language
- Software Testing & Verification
- AI & Heterogeneous Computing

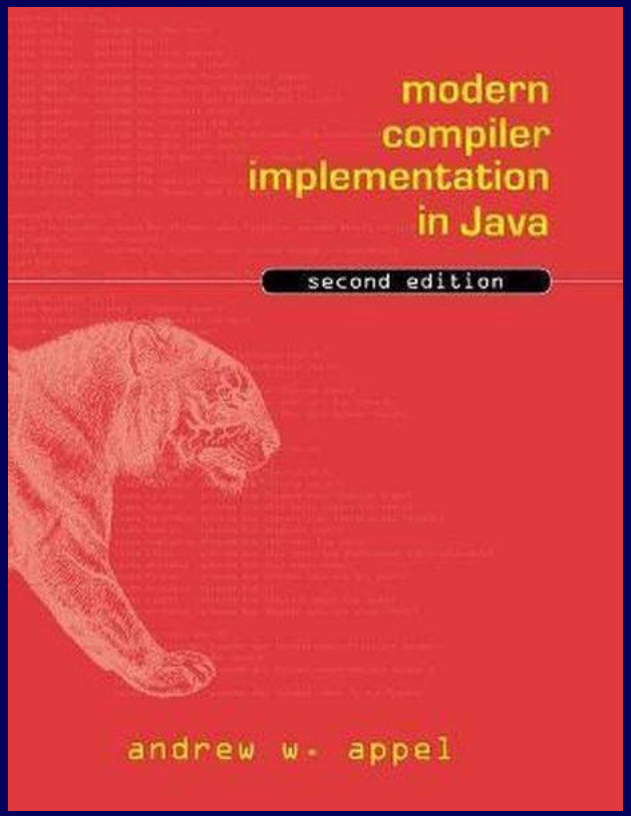
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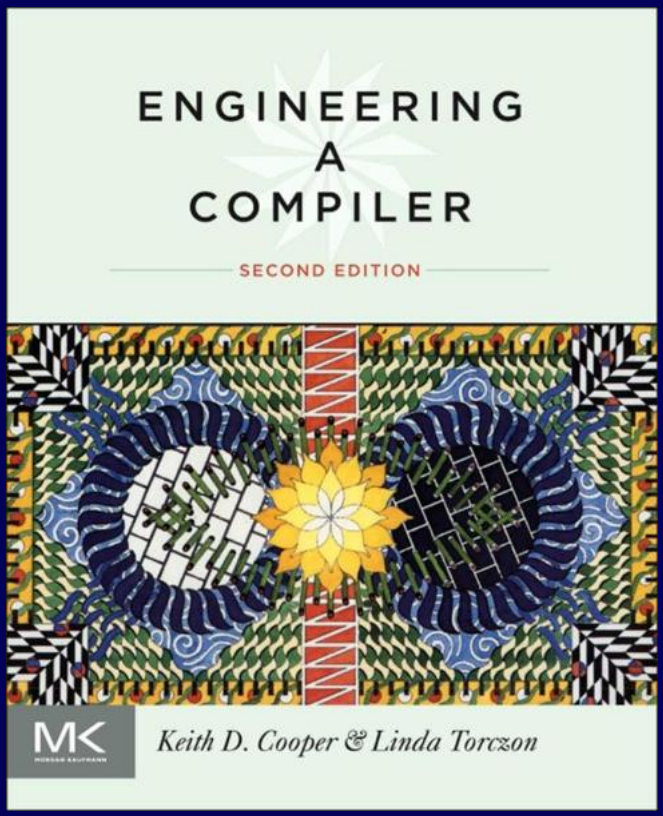




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Thank You

